ECON 0100 | HW C3 | Corrective Policies

Due: Sunday, Oct. 18

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Q1 | Malodorous Byproducts

Butterbeer is beloved by Hogwarts students and teachers alike and is widely available at wizarding shops. But the beverage has the negative effect of generating displeasing odors from its imbibers. Preferences for butterbeer and its marginal cost can be represented by the following relationships.

$$D : P_B = 100 - \frac{1}{2}Q_D \qquad S : P_S = 20 + \frac{1}{2}Q_S$$

Prices are in galleons and quantity is in bottles. The odor imposes a cost of 20 galleons per bottle on everyone nearby. Using your economic policy advising credentials, use a graph and some algebra to offer a remedy to the malodorous byproduct that the Ministry could implement.

Policy (tax or subsidy): _____ Size per bottle: _____

Market Quantity ($\hat{Q}$): _____ Quantity with Your Policy: _____

Q2 | Dragon Pox

Dragon Pox is a virus that has spread widely throughout the wizarding world. Gorsemoor's Dose prevents a witch or wizard from catching it, meaning they also don't pass Dragon Pox on to anyone else. The marginal private benefit of a dose is $P = 24 - Q$, and supply, which is both the private and the social cost, is $P = 4 + Q$. Prices are in galleons and quantity is in doses. Each dose also gives a benefit of 6 galleons to everyone else, since they are less likely to catch Dragon Pox.

Offer a remedy the Ministry could implement to correct the market failure. And conclude with a description of the effects your policy would have on the market for Gorsemoor's Dose.

Policy (tax or subsidy): _____ Size per dose: _____

Market Quantity ($\hat{Q}$): _____ Quantity with Your Policy: _____

How does your policy bring the market to the socially efficient quantity?

a) It lowers the cost of brewing each dose, so sellers offer fewer.

b) It raises the price buyers pay, so fewer witches and wizards drink it.

c) Buyers now receive the benefit to everyone else on every dose, so their benefit plus the subsidy matches marginal social benefit, and the market settles where marginal social benefit equals marginal cost.

d) It sets a price floor, so sellers earn more on every dose.